GeoNews — Week 31/2026

A weekly digest on Quaternary palaeoclimate research, East African rift dynamics, palaeohydrology and time-series analysis · 27 July 2026

> **What is GeoNews?** An **automatically generated weekly digest**. Every Monday a research workflow screens the publications of the past one to two weeks along with funding, policy and community sources, scores the hits against a fixed set of criteria, and assembles this issue from the result.
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> The thematic profile is **aligned with the research interests of Markus L. Fischer** (Palaeoclimate Dynamics group, University of Potsdam): Quaternary palaeoclimate and palaeoenvironmental reconstruction, the East African Rift, the African Humid Period and human–environment coupling, lake-level and water-balance modelling, palaeofloods and fluvial geomorphology, ICDP drill cores, and linear and nonlinear time-series analysis. Anyone working in these fields will probably find something useful here — it is explicitly not a complete survey of Quaternary science.
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> Selection, scoring and summaries are produced by machine and reviewed before publication. Details may contain errors; where in doubt, the linked original source governs.

Each paper receives a score from 0 to 100 across five criteria: thematic fit (40), methodological fit (25), impact/venue (15), applicability (10) and recency (10). Categories: 🔴 75 and above, must-read · 🟡 50–74, relevant · 🟢 30–49, on the radar. The score measures proximity to this newsletter's profile and sets reading priority; it is not a quality judgement on the individual paper.

In brief

Eight relevant papers, four on the radar, plus ten title-level records from the Elsevier core journals — no must-read this time. Crossref access was blocked during the Monday run but could be restored the same day; the core coverage was then pulled in full (QSR, EPSL, GPC, P³, Nature Geoscience, Boreas, JQS, The Holocene, Quaternary Research, Geomorphology, Catena, Earth-Science Reviews). One structural gap remains: Elsevier deposits no abstracts with Crossref, and ScienceDirect is not reachable from this compute environment — those hits therefore sit in a separate block, with title, journal, date and DOI, but without content annotation and without a score.

Closest to the core is an **EGUsphere preprint on the abruptness of the end of the Green Sahara** (Claussen et al.): a quantitative, changepoint-based analysis combining an MPI-ESM transient simulation with dust and pollen proxies that resolves the long-standing "collapse versus gradual decline" debate — gradual desert expansion at the large scale, but abrupt, step-like transitions at the ecosystem level. The preprint sits outside the 7–14-day window (posted 20 May 2026) but is currently under active review (author replies 15 July 2026) and is therefore included. From the actual window come a **Nature-family paper on mid- to late Holocene cooling and its influence on agriculture on the Tibetan Plateau** (Monte-Carlo-based temperature reconstruction, human–environment coupling) and a **Climate of the Past paper that couples cosmogenic multi-nuclide dating with proglacial lake sediments** to reconstruct the Holocene history of a small Scandinavian ice cap — methodologically the most interesting combination of the week. Two further papers with a direct profile fit come from the Crossref follow-up: a **quantitative multiproxy reconstruction of the last deglaciation from the Korean palaeolake Gonggeomji**, coupling vegetation, fire and hydroclimate in a single record, and a **CEE paper applying neural networks to a physically defined circulation index of the West African monsoon**, weighting models by process fidelity rather than by hit rate.

On funding, the focus is unchanged: the dedicated Horizon palaeoclimate call **HORIZON-CL5-2027-01-D1-08** (opening 17 November 2026, deadline 4 March 2027), the **ERC Starting Grant 2027** (15 October 2026) and the **PAGES applications** (11 September 2026). Near-term: the **AGU abstract deadline of 5 August 2026**. On research policy the federal budget has firmed up: the cabinet already adopted the 2027 draft **on 6 July 2026** — the research ministry's core budget grows only about 0.7 percent nominally, with real increases almost exclusively in the special fund.

Part A — Papers of the week

Papers were captured with a `created` or online date from 13 July 2026 onwards, plus one preprint currently under review that lies outside this window (clearly flagged). DOIs already listed in Week 30 are excluded. Every scored entry in the 🟡 and 🟢 categories rests on an abstract or short summary; the closing ⚬ block lists papers for which no abstract was accessible — without a score and without any claim about their content. There is no 🔴 must-read this week.

🟡 Relevant (50–74)

1. How abruptly did the Holocene Green Sahara end? · *EGUsphere (preprint, under review at Climate of the Past)*, 2026 · **Score 74** — ⚠ *Preprint, outside the window (posted 20 May 2026), currently under active review (author replies 15 July 2026)* A quantitative reassessment of the termination of the African Humid Period in the Sahara. The authors characterise the abruptness of vegetation change in a transient simulation of the MPI Earth System model and in proxy data (dust flux, pollen) over the last 8000 years using a changepoint approach. Central finding: in a northwest–southeast oriented strip, step-like transitions in bare surface occur on average about four times faster than the approximately linear trend in orbital forcing; the reconstructed dust flux is even more abrupt, while the pollen records are largely more gradual. The work supports the view that there was no "collapse" of the Green Sahara and that gradual and abrupt signals in different archives are not contradictory. Doubly relevant here: the African Humid Period on one hand, and methodologically the quantification of abruptness in time series on the other — the same question addressed for the AHP termination using early-warning signals. *Theme 40 · Method 20 · Venue 4 · Applicability 8 · Recency 2 (posted 20 May 2026)* · DOI: 10.5194/egusphere-2026-2587

2. Mid-late Holocene cooling shaped agricultural transformation on the Tibetan Plateau · *Communications Earth & Environment*, 2026 · **Score 72** A reconstruction of mid- to late Holocene temperature variability from newly generated and published proxy data, with Monte Carlo simulations to handle uncertainty. Result: cooling drives a shift from thermophilic millet cultivation to a cold-resilient wheat–barley system; the *rate* of temperature change — not only its magnitude — controls agricultural stability, with phases of heightened variability coinciding with declines in crop productivity. Relevant as a cleanly quantified example of human–environment coupling in the Holocene, and conceptually transferable to questions of environmental thresholds in settlement history of the kind also posed for East African contexts.

Theme 25 · Method 18 · Venue 11 · Applicability 8 · Recency 10 (created 27 Jul 2026) · DOI: 10.1038/s43247-026-03838-9

3. Riukojietna, a small low-altitude ice cap that may have persisted through the Holocene: evidence from combining cosmogenic multi-nuclide dating and lacustrine sediment records · *Climate of the Past*, 2026 · **Score 68** A reconstruction of the Holocene history of a flat, low-altitude plateau ice cap in northern Sweden. It combines cosmogenic multi-nuclide measurements in bedrock (in situ ^{14}C, ^{10}Be, ^{26}Al) — which constrain exposure and burial phases — with indirect evidence of glacial activity from proglacial lake sediments and a subsequent forward-modelling step. Finding: the ice cap survived the Holocene Thermal Maximum (ca. 8–5 ka), contrary to earlier suggestions of a complete disappearance of Scandinavian glaciers, probably owing to increased precipitation. Methodologically the most interesting paper of the week: coupling short- and long-lived cosmogenic nuclides with lake-sediment chronology is an architecture that transfers directly to other high-latitude and alpine Quaternary archives. *Theme 25 · Method 17 · Venue 11 · Applicability 5 · Recency 10 (created 27 Jul 2026)* · DOI: 10.5194/cp-22-1363-2026

4. Last deglaciation hydroclimatic environment on the central inland of the Korean Peninsula based on pollen and microscopic charcoal records · *Quaternary Research*, 2026 · **Score 66** A multiproxy reconstruction from the Gonggeomji palaeolake in central South Korea covering the last deglaciation (ca. 17–10 ka). It combines pollen-based quantitative estimates of mean annual temperature and precipitation with palynomorph assemblages, sedimentary charcoal and lithological data. Finding: during Heinrich Stadial 1, cold and dry conditions — attributed by the authors to a weakened Atlantic overturning circulation and an intensified East Asian winter monsoon — sustained boreal conifer forests, reduced vegetation cover, and increased fire frequency and soil erosion. The Bølling–Allerød brought rapid warming, higher humidity and warm-temperate deciduous forests; the Younger Dryas a brief reversal with renewed fire activity while moisture stayed moderate, with post-fire recovery following a classical successional sequence. Relevant as a palaeolake archive with quantitative climate reconstruction and explicitly coupled vegetation–fire–hydroclimate dynamics — the same analytical architecture applied to rift lake sediments, and a clean example of high- and low-latitude forcings converging in a single record. *Theme 25 · Method 18 · Venue 8 · Applicability 5 · Recency 10 (created 20 Jul 2026)* · DOI: 10.1017/qua.2026.10099

5. Constraining high-emission future West African monsoon with physics-weighted deep learning ensembles · *Communications Earth & Environment*, 2026 · **Score 65** Physics-guided neural networks learn the relationship between the five sea-level-pressure boxes of the Sahelian monsoon ocean-pressure index (SMOPI) and June–September monsoon rainfall — trained individually on eleven CMIP6 models, then all forced with the same JRA-55 reanalysis. The physical realism of the learned teleconnections then serves as the weight in the ensemble. Result: end-of-century inter-model spread falls by 20 percent among the high-skill models, 33 percent among the remainder and 30 percent across the full ensemble, without collapsing ensemble diversity. Methodologically notable: the authors show explicitly that a high index correlation alone does not establish correct physics — a model can be right for the wrong reasons. For this profile the architecture matters more than the projections: non-linear learning on a physically defined circulation index plus weighting by process fidelity rather than hit rate — the question of which models to weight for palaeo-monsoon phases is the same one. *Theme 25 · Method 13 · Venue 11 · Applicability 6 · Recency 10 (created 21 Jul 2026)* · DOI: 10.1038/s43247-026-03839-8

6. Pleistocene benthic foraminifera bioevents in the Central Arctic Ocean: stratigraphic and paleoceanographic implications · *Climate of the Past*, 2026 · **Score 62** A biostratigraphic subdivision of the central Arctic Pleistocene based on benthic foraminiferal bioevents, with palaeoceanographic interpretation. Relevant as a Quaternary multiproxy/ biostratigraphy reference: constructing robust event stratigraphies in hard-to-date marine sequences is methodologically related to the question of how to anchor event markers in terrestrial cores lacking a continuous absolute chronology. *Theme 25 · Method 13 · Venue 11 · Applicability 3 · Recency 10 (created 21 Jul 2026)* · DOI: 10.5194/cp-22-1305-2026

7. Climatic, tectonic and landslide-dam signals preserved in alluvial terraces of the Naryn Basin, Kyrgyzstan · *EGUsphere (preprint, under review at Earth Surface Dynamics)*, 2026 · **Score 60** — ⚠ *Preprint* Mapping, dating and modelling of fluvial terraces on the Naryn in the Central Asian Tien Shan. The work separates three superimposed signals — climatic terrace formation, tectonic uplift, and the rapid drainage of a landslide-dammed lake — and finds a spatially limited outburst-flood signal against a regionally effective climatic terrace driver. The same methodological core question arises in palaeoflood research: how to distinguish climatically driven from non-climatic (here: outburst-related) extreme discharges in a geomorphological archive — a direct contrasting case to climatically driven flood generation. *Theme 25 · Method 17 · Venue 4 · Applicability 8 · Recency 6 (posted 14 Jul 2026)* · DOI: 10.5194/egusphere-2026-3489

8. A multiproxy reconstruction of changes in sea ice and primary productivity at IODP Site U1339 (Umnak Plateau, Bering Sea) during Marine Isotope Stage 11 · *Quaternary Research*, 2026 · **Score 59** A multiproxy record from sedimentology, diatoms and organic geochemistry at IODP Site U1339 in the southeastern Bering Sea, covering sea ice and primary productivity from the end of MIS 12 to the beginning of MIS 10 (430–368 ka). Finding: in late MIS 12 the site had extensive seasonal sea ice with spring concentrations near 100 percent; during deglaciation it became year-round ice-free, yet sea ice readvanced during the interglacial optimum — unlike the Holocene, when no sea ice occurs at the Umnak Plateau. Productivity does not stay high through the interglacial but declines independently of sea ice and upwelling, possibly through increasing nutrient limitation. Relevant as a reference for MIS 11 — the interglacial routinely invoked as a Holocene analogue — and as evidence that proxies within one interglacial can diverge rather than track a common climate signal. *Theme 25 · Method 13 · Venue 8 · Applicability 3 · Recency 10 (created 17 Jul 2026)* · DOI: 10.1017/qua.2026.10094

● On the radar

- **Ice core nitrogen isotopes archive dramatic changes in West Antarctic Ice Sheet thinning** · *Climate of the Past* · **45** ● · A new nitrogen-isotope proxy in ice cores to reconstruct thinning of the West Antarctic Ice Sheet; methodologically interesting, thematically at the edge of the profile. DOI: 10.5194/cp-22-1291-2026
- **Sediment storage and routing in bedrock canyons** · *Earth Surface Dynamics* · **41** ● · Repeat channel-bed surveys quantify sediment storage in deep bedrock canyons (vertical changes up to 15 m); transferable to interpreting sediment supply and storage processes in fluvial systems. DOI: 10.5194/esurf-14-553-2026
- **ImageGrains 2.0: Improved precision and generalization for grain segmentation** · *Earth Surface Dynamics* · **40** ● · An updated image-segmentation framework for automatic grain-size measurement; usable as an open tool for sedimentological image and scan analysis. DOI: 10.5194/esurf-14-527-2026

- **The geomorphology of burrowing bioturbation in the Central French Pyrenees** · *EGUsphere (preprint, under review at Earth Surface Dynamics)* · **35** ● · Burrowing mammals as a geomorphological factor in mountain grasslands; a peripheral topic, noted as an example of bioturbation effects on surface processes. DOI: 10.5194/egusphere-2026-2910

○ Screened without an abstract — Elsevier core journals

These ten papers come from the Elsevier journals in the profile and fall inside the window. Crossref records them with title, journal, DOI and `created` date, but deposits no abstracts for Elsevier; ScienceDirect is not retrievable from this compute environment, and the articles are days old and not yet in search indices. They are therefore listed **without a score and without content annotation** — the note states only the thematic connection that follows from the title. What the papers actually say is decided by the full text.

Palaeofloods and fluvial extreme events

- **Post-glacial landforms dictate fluvial sediment dynamics during extreme floods: Morphological adjustments and flood hazard implications** · *Geomorphology*, created 21 Jul 2026 · Thematic link per title: sediment dynamics and morphological adjustment during extreme floods in glacially preconditioned valley systems. DOI: 10.1016/j.geomorph.2026.110457
- **Quantitative reconstruction of the damming and outburst processes of the Temi Palaeolandslide in the Upper Jinsha River, Southeastern Tibetan Plateau, China** · *Palaeogeography, Palaeoclimatology, Palaeoecology*, created 16 Jul 2026 · Thematic link per title: quantitative reconstruction of damming and outburst — a contrasting case to climatically driven extreme discharge, thematically parallel to the Naryn paper above. DOI: 10.1016/j.palaeo.2026.114068

Lake sediments and palaeohydrology

- **Millennial-scale carbon-source changes revealed by sedimentary radiocarbon in Arctic lakes** · *Quaternary Science Reviews*, created 21 Jul 2026 · Thematic link per title: carbon sources in lake sediments via radiocarbon — touches on reservoir and age effects of the kind encountered in lake chronologies. DOI: 10.1016/j.quascirev.2026.110121
- **Evidence of a Weichselian MIS 3 proglacial lake south of Dogger Bank (southern North Sea)** · *Quaternary Science Reviews*, created 26 Jul 2026 · Thematic link per title: evidence for a proglacial lake during MIS 3. DOI: 10.1016/j.quascirev.2026.110190
- **Millennial cold–dry events during marine isotope stage 11a based on sedimentary records from the Heqing Basin and their global climatic dynamical responses** · *Global and Planetary Change*, created 23 Jul 2026 · Thematic link per title: millennial cold–dry events in a lake-basin record, MIS 11a. DOI: 10.1016/j.gloplacha.2026.105634
- **Mid- to late holocene palaeoenvironmental and palaeoecological dynamics in Lake Azaplı (Gölbaşı Basin, SE Türkiye): Insights from pollen and non-pollen palynomorph records** · *Palaeogeography, Palaeoclimatology, Palaeoecology*, created 22 Jul 2026 · Thematic link per title: pollen and non-pollen palynomorphs from a lake archive, mid to late Holocene. DOI: 10.1016/j.palaeo.2026.114085

Proxies, chronology and sediment budgets

- **Assessing GDGT-derived (sub)surface temperature proxies in Arctic fjord sediments: A 14 kyr record from the Kongsfjorden Trough, Svalbard** · *Palaeogeography, Palaeoclimatology, Palaeoecology*, created 23 Jul 2026 · Thematic link per title: evaluation of organic-geochemical temperature proxies against a 14 ka record. DOI: 10.1016/j.palaeo.2026.114070

- **Cave-based chronology of valley incision in the glaciated Northern Carpathians** · *Earth and Planetary Science Letters*, created 23 Jul 2026 · Thematic link per title: dating valley incision via cave levels. DOI: 10.1016/j.epsl.2026.120229
- **Disentangling diffuse vs. concentrated erosion in total soil exports over the Holocene: combining erosion models, sediment proxies, remote sensing and geophysics** · *Catena*, created 25 Jul 2026 · Thematic link per title: partitioning the Holocene sediment budget across erosion processes, combining models and proxies. DOI: 10.1016/j.catena.2026.110450
- **Rivers in motion: a wavelet-based remote sensing approach to spatio-temporal hydro-geomorphic analysis** · *Catena*, created 22 Jul 2026 · Thematic link per title: wavelet analysis as a space–time method applied to fluvial remote-sensing data. DOI: 10.1016/j.catena.2026.110451

Part B — Funding & calls

Horizon Europe — "Palaeoclimate science for a better understanding of Earth system dynamics" (HORIZON-CL5-2027-01-D1-08). Opening **17 November 2026**, deadline **4 March 2027**, Research and Innovation Action, budget around **€14 million for roughly two projects**, so about €7 million per consortium. The stated aim is robust information on palaeoclimate states and events outside the range of variability recorded over the past centuries, to guide the development of Earth system models. With two funded projects Europe-wide the competition is severe, and consortium discussions belong early in the timeline. Sources: topic page · CL5 Work Programme 2026–2027 (PDF).

ERC Starting Grant 2027 — deadline 15 October 2026. Step 1 results in April 2027, interviews and Step 2 results in August 2027. Eligibility: PhD defended no more than ten years before 1 January 2027, extendable for parental and care leave. New from the 2027 calls: at most one Starting and one Consolidator Grant per career, plus stricter application rules. The **ERC Consolidator 2027** is expected in January 2027. Sources: ERC — Applying in the 2027 competitions · ERC — Changes to the 2026 and 2027 Work Programmes.

PAGES — working group and workshop support, deadline 11 September 2026. The next application round covers workshops, meetings, new working groups and extensions of existing groups. Ahead of the Horizon palaeoclimate call this is a convenient instrument for forming a visible consortium. Also: the **PAGES Early-Career Award 2027**, applications close **29 September 2026**. Sources: PAGES — Workshop/meeting support · PAGES Newsletter July 2026.

ICDP — proposal cycle. The regular annual deadline for full, pre- and workshop proposals is **15 January**; the 2027 cycle is likely to follow the same pattern, though no formal announcement has appeared. Workshop calls run through the year. Source: ICDP — Proposals.

DFG — changed conditions. No new palaeoclimate-specific call. Relevant for planning: the **programme allowance rises from 22 to 25 percent from 2027**; the next **Priority Programme application round** takes place only in **late 2026**, with funding starting in **2028**. Source: Forschung & Lehre.

Part C — Research policy (Germany/EU)

Federal budget 2027 — cabinet decision already on 6 July 2026. Contrary to what was still expected the week before, the cabinet adopted the government draft for the 2027 federal budget as early as 6 July 2026. The research ministry's (BMFTR) core budget stands at around **€22 billion** — a nominal increase of only about **0.7 percent** over 2026. Around €764 million comes from the Climate and Transformation Fund (plus about €45 million), and the R&D investments earmarked in the special fund for infrastructure and climate neutrality rise by roughly 60 percent in 2027 over 2026. As interpretation rather than fact: the regular budget thus stagnates in real terms given current pay and materials cost increases, while the growth flows almost exclusively into the time-limited special fund — adding to the operational pressure on the funding organisations. Sources: Forschung & Lehre — only a slim increase for the BMFTR core budget · Table.Briefings — Budget 2027 · Federal Ministry of Finance — 2027 government draft.

DFG cuts at the postdoc level. Facing high collective-agreement pay rises and increased equipment and consumables costs that the guaranteed annual funding increments do not cover, the DFG has postponed the funding start for new Research Training Group applications and **suspended approval of postdoc positions**. For postdoc career paths in the geosciences this removes one funding route for the time being, increasing pressure on individual grants and EU money. Source: Fakultätentag Informatik.

WissZeitVG — ministry draft published. The core of the draft is **minimum contract durations of three years for first contracts of doctoral researchers and two years for postdocs** in the respective qualification phase. The maximum fixed-term duration remains six years per phase; the "four plus two" arrangement from the failed previous draft is off the table. Unfavourable for the postdoc phase: the previous carry-over provision is removed, under which unused fixed-term time from the doctoral phase could be transferred. Practical consequence: anyone carrying unused time from the doctoral phase should check how the draft affects their personal fixed-term account. Sources: Wiarda blog · Forschung & Lehre.

Part D — Community & deadlines

- **AGU Fall Meeting 2026** — **7–11 December 2026 in San Francisco**; abstract deadline **5 August 2026** (23:59 EDT). Source: AGU.
- **PAGES working group and workshop applications** — **11 September 2026** · **PAGES Early-Career Award 2027** — **29 September 2026**.
- **DEUQUA conference 2026** — **28 September to 2 October 2026 in Freiburg im Breisgau**, with full-day excursions on 28 September and 2 October. The Göttingen meeting originally planned for 2026 has been postponed to 2028. Source: DEUQUA.
- **ERC Starting Grant 2027** — deadline **15 October 2026**.
- **Horizon CL5 palaeoclimate call** — opening **17 November 2026**, deadline **4 March 2027**.
- **EGU General Assembly 2027** — **4 to 9 April 2027 in Vienna**, hybrid. The abstract deadline is not yet published; experience suggests mid-January. Source: EGU.
- **ICDP** — next regular proposal deadline expected **15 January 2027**, following previous years' pattern and not yet confirmed.

Method & sources

Paper screening normally uses the Crossref REST API with topic- and ISSN-based queries and the `from-created-date` filter. DOIs already listed in an earlier issue are not repeated. Funding, research policy and deadlines rest on the sources linked in Parts B to D.

How this issue came about. The Monday run initially proceeded without Crossref: API access from the compute environment was blocked (proxy 403) and the intended browser fallback unavailable. The first version therefore rested solely on the recent-article listings of the open-access journals (*Climate of the Past*, *Earth Surface Dynamics*, *Geochronology*, *EGUsphere*) and the article list of *Communications Earth & Environment*. Crossref became reachable again the same day; the core coverage was then pulled journal by journal via ISSN — *Quaternary Science Reviews*, *Earth and Planetary Science Letters*, *Global and Planetary Change*, *Palaeogeography, Palaeoclimatology, Palaeoecology*, *Nature Geoscience*, *Communications Earth & Environment*, *Boreas*, *Journal of Quaternary Science*, *Quaternary Geochronology*, *The Holocene*, *Quaternary Research*, *Geomorphology*, *Catena*, *Earth-Science Reviews*, *Journal of Human Evolution* — and the issue extended with the hits from that follow-up. Coverage is therefore back to the level of a normal weekly run.

What remains open. Elsevier deposits no abstracts with Crossref. The usual workaround via the publisher page was unavailable: ScienceDirect responds to this environment with a bot check, and the papers are days old and not yet indexed by search engines. Rather than inferring results from titles, those ten papers sit in a separate, clearly marked block without a score and without any claim about their content. For the scored entries the standard is unchanged: every substantive statement rests on an abstract that was inspected. The EGUsphere preprint layer was additionally included, with every preprint flagged as such. DOIs already listed in Week 30 are not repeated — in this window that includes, among others, the Mojave palaeohydrology paper, the Dadu outburst floods and the Nature Geoscience paper on core imaging.

GeoNews is an automatically generated digest published by Dr Markus L. Fischer, Palaeoclimate Dynamics group, University of Potsdam. The thematic profile follows his research interests; selection is made by a rule-based workflow, not an editorial team. The digest is an independent literature review with no affiliation to the publishers or funding bodies mentioned, and does not constitute a recommendation or advice. Corrections and pointers are welcome.